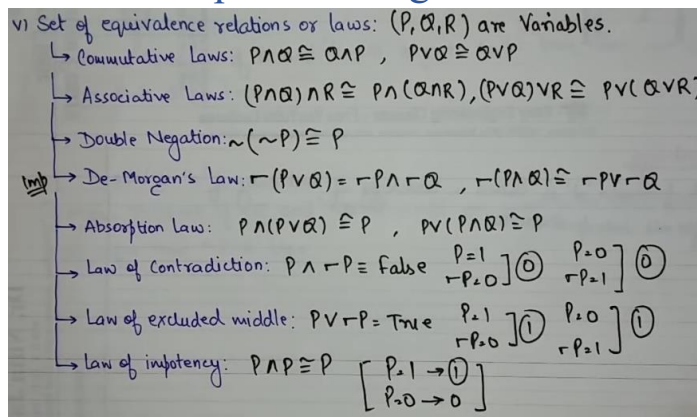


MODULE 4: PROPOSITION LOGIC

Contents

1. Laws in Proposition Logic.....	1
2. Types of Inference rules	2
3. Inference Rules	5
4. Well-formed formula to Predicate Logic.....	7
5. Steps to Convert Logical Statement to CNF.....	9
6. Resolution.....	11
7. Bayesian Networks	15
8. Dempster-Shafer Theory (DST).....	18
9. Fuzzy Sets.....	21
10. Fuzzy Logic	21
a. Fuzzy logic Operations.....	22
b. Algebraic sum & product, bounded sum & difference.....	24
c. Cartesian Product	25
d. Max Min composition & Max product composition.....	25
e. Membership value for triangles using inference approach.....	26
11. Defuzzification.....	27

Laws in Proposition Logic



Example: Using First-Order Logic for Reasoning

To illustrate the use of FOL in reasoning, consider the following knowledge base:

1. $\forall x (Cat(x) \rightarrow Mammal(x))$ (All cats are mammals)
2. $\forall x (Mammal(x) \rightarrow Animal(x))$ (All mammals are animals)
3. $Cat(Tom)$ (Tom is a cat)

From these statements, we can infer:

1. $Mammal(Tom)$ (Since Tom is a cat, and all cats are mammals)
2. $Animal(Tom)$ (Since Tom is a mammal, and all mammals are animals)

These inferences demonstrate how FOL can be used to derive new knowledge from existing facts. The ability to reason logically about relationships and properties is a key strength of FOL in AI.

Q The CORRECT formula for the sentence, "not all rainy days are cold" is

a) $\forall_d (\text{Rainy}(d) \wedge \sim \text{Cold}(d))$ b) $\forall_d (\sim \text{Rainy}(d) \rightarrow \text{Cold}(d))$
 c) $\exists_d (\sim \text{Rainy}(d) \rightarrow \text{Cold}(d))$ d) $\exists_d (\text{Rainy}(d) \wedge \sim \text{Cold}(d))$

Q Consider the statement "Not all that glitters is gold"
 Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

a) $\forall_x: \text{glitters}(x) \Rightarrow \sim \text{gold}(x)$ c) $\exists_x: \text{gold}(x) \wedge \sim \text{glitters}(x)$
 b) $\forall_x: \text{gold}(x) \Rightarrow \text{glitters}(x)$ d) $\exists_x: \text{glitters}(x) \wedge \sim \text{gold}(x)$

Q What is the logical translation of the following statement?
 "None of my friends are perfect."

A) $\exists_x (F(x) \wedge \sim P(x))$ B) $\exists_x (\sim F(x) \wedge P(x))$
 C) $\exists_x (\sim F(x) \wedge \sim P(x))$ D) $\sim \exists_x (F(x) \wedge P(x))$

Q Which one of the following is the most appropriate logical formula to represent the statement? "Gold and silver ornaments are precious". The following notations are used:
 G(x): x is a gold ornament
 S(x): x is a silver ornament
 P(x): x is precious

(A) $\forall_x (P(x) \rightarrow (G(x) \wedge S(x)))$
 (B) $\forall_x ((G(x) \wedge S(x)) \rightarrow P(x))$
 (C) $\exists_x ((G(x) \wedge S(x)) \rightarrow P(x))$
 (D) $\forall_x ((G(x) \vee S(x)) \rightarrow P(x))$

Q Which one of the first order predicate calculus statements given below correctly express the following English statement? "Tigers and lions attack if they are hungry or threatened"

a) $\forall_x ((\text{tiger}(x) \wedge \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x))$
 b) $\forall_x ((\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \wedge \text{attacks}(x))$
 c) $\forall_x ((\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \text{attacks}(x) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)))$
 d) $\forall_x ((\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x))$

Types of Inference rules

1. Modus Ponens: The Modus Ponens rule is one of the most important rules of inference, and it states that if P and $P \rightarrow Q$ is true, then we can infer that Q will be true. It can be represented as:

Notation for Modus ponens:
$$\frac{P \rightarrow Q, P}{\therefore Q}$$

Example:

Statement-1: "If I am sleepy then I go to bed" $\Rightarrow P \rightarrow Q$

Statement-2: "I am sleepy" $\Rightarrow P$

Conclusion: "I go to bed." $\Rightarrow Q$.

Hence, we can say that, if $P \rightarrow Q$ is true and P is true then Q will be true.

P	Q	$P \rightarrow Q$
0	0	0
0	1	1
1	0	0
1	1	1

2. Modus Tollens: The Modus Tollens rule state that if $P \rightarrow Q$ is true and $\neg Q$ is true, then $\neg P$ will also true. It can be represented as:

Notation for Modus Tollens:
$$\frac{P \rightarrow Q, \sim Q}{\sim P}$$

Statement-1: "If I am sleepy then I go to bed" $\Rightarrow P \rightarrow Q$

Statement-2: "I do not go to the bed." $\Rightarrow \sim Q$

Statement-3: Which infers that "I am not sleepy" $\Rightarrow \sim P$

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	0
1	1	0	0	1

3. Hypothetical Syllogism: The Hypothetical Syllogism rule state that if $P \rightarrow R$ is true whenever $P \rightarrow Q$ is true, and $Q \rightarrow R$ is true. It can be represented as the following notation:

Example:

Statement-1: If you have my home key then you can unlock my home. $P \rightarrow Q$

Statement-2: If you can unlock my home then you can take my money. $Q \rightarrow R$

Conclusion: If you have my home key then you can take my money. $P \rightarrow R$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	1

4. Disjunctive Syllogism: The Disjunctive syllogism rule state that if $P \vee Q$ is true, and $\neg P$ is true, then Q will be true. It can be represented as:

Notation of Disjunctive syllogism:
$$\frac{P \vee Q, \neg P}{Q}$$

Example:

Statement-1: Today is Sunday or Monday. $\Rightarrow P \vee Q$

Statement-2: Today is not Sunday. $\Rightarrow \neg P$

Conclusion: Today is Monday. $\Rightarrow Q$

P	Q	$\neg P$	$P \vee Q$
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1

5. Addition: The Addition rule is one the common inference rule, and it states that If P is true, then $P \vee Q$ will be true.

Notation of Addition:
$$\frac{P}{P \vee Q}$$

Example:

Statement: I have a vanilla ice-cream. $\implies P$

Statement-2: I have Chocolate ice-cream.

Conclusion: I have vanilla or chocolate ice-cream. $\implies (P \vee Q)$

P	Q	$P \vee Q$
0	0	0
1	0	1
0	1	1
1	1	1

6. Simplification: The simplification rule state that if $P \wedge Q$ is true, then **Q or P** will also be true. It can be represented as:

Notation of Simplification rule: $\frac{P \wedge Q}{Q}$ Or $\frac{P \wedge Q}{P}$

Proof by Truth-Table:

P	Q	$P \wedge Q$
0	0	0
1	0	0
0	1	0
1	1	1

7. Resolution: The Resolution rule state that if $P \vee Q$ and $\neg P \vee R$ is true, then $Q \vee R$ will also be true. **It can be represented as**

Notation of Resolution $\frac{P \vee Q, \neg P \vee R}{Q \vee R}$

Proof by Truth-Table:

P	$\neg P$	Q	R	$P \vee Q$	$\neg P \vee R$	$Q \vee R$
0	1	0	0	0	1	0
0	1	0	1	0	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	0	1
1	0	1	1	1	1	1

Inference Rules

Inference Rules

① Modus Ponens

$$\frac{P \rightarrow Q \quad P}{Q}$$

② Modus Tollens

$$\frac{P \rightarrow Q \quad \sim Q}{\sim P}$$

③ Hypothetical Syllogism: $P \rightarrow Q$
 $Q \rightarrow R$
 $\hline P \rightarrow R$

④ Disjunctive Syllogism: $P \vee Q$
 $\sim P$
 $\hline Q$

⑤ OR Introduction
~~Addition~~ $\frac{P}{P \vee Q} \text{ or } \frac{Q}{P \vee Q}$

⑥ Simplification Rule

$$\frac{P \wedge Q}{P} \text{ or } \frac{P \wedge Q}{Q}$$

⑦ Resolution

$$\frac{P \vee Q \quad \sim P \vee R}{Q \vee R}$$

① $(A \vee B) \wedge C \rightarrow E$
 $D \wedge F \rightarrow A$
 $\sim E \vee D$
 $B \wedge C$
 F
 $\hline A$

Ans: $\frac{B \wedge C}{B} + \frac{B \wedge C}{C} \{ \text{Simplification} \}$

$\therefore B \wedge C$ are true.

② $\frac{B}{(A \vee B)}$ ~~OR~~ $\{ \text{OR introduction} \}$

③ we see $(A \vee B)$ true & C is true.

$\therefore (A \vee B) \wedge C$ is true.

④ $(A \vee B) \wedge C \rightarrow E$
 $(A \vee B) \wedge C$
 $\hline E$

⑤ $\sim E \vee D \equiv F \rightarrow D$ $\{ \text{Modus Ponens} \}$
 $\frac{F}{D}$

⑥ we have F true & D true $\therefore (D \wedge F)$ is true.
 $D \wedge F \rightarrow A$
 $\frac{D \wedge F}{A} \{ \text{Modus Ponens} \}$

$$\begin{array}{l|l} \textcircled{2} (A \vee \neg C) \rightarrow B & \textcircled{1} \neg C \vee D \equiv C \rightarrow D \quad \{ \text{modus ponens} \} \\ \neg C \vee D & \frac{C}{D} \\ D \rightarrow A & \\ \hline C & \\ \hline A \wedge B & \end{array}$$

$$\textcircled{2} \frac{A}{A \vee \neg C} \quad \{ \text{OR Introduction} \}$$

$$\begin{array}{l} \textcircled{1} (A \vee \neg C) \rightarrow B \\ (A \vee \neg C) \\ \hline B \end{array}$$

$$\begin{array}{l} \textcircled{1} A \\ B \\ \hline A \wedge B \end{array}$$

① All Birds can fly: $\forall x \text{ Bird}(x) \rightarrow \text{fly}(x)$

② Every man respects his parent: $\forall x \text{ Man}(x) \rightarrow \text{respects}(x, \text{parent})$

③ Some boys play cricket: $\exists x \text{ Boys}(x) \rightarrow \text{play}(x, \text{cricket})$

④ Not all students like both Maths & Science.

$\neg \forall x [\text{Student}(x) \rightarrow (\text{like}(x, \text{Maths}) \wedge \text{like}(x, \text{Science}))]$

⑤ Only one student failed in Mathematics.

$\exists (x) [\text{Student}(x) \rightarrow \text{failed}(x, \text{Maths}) \wedge \forall (y) [\neg (x=y) \wedge \text{Student}(y) \rightarrow \neg \text{failed}(y, \text{Maths})]]$

⑥ Not all that glitters is gold: $\neg \forall x \text{ Glitters}(x) \rightarrow \text{gold}(x)$

⑦ Some real nos are rational: $\exists x \text{ Real}(x) \rightarrow \text{Rational}(x)$

Well-formed formula to Predicate Logic

Q.1) Sentences to Well formed Formulas in Predicate Logic

① Marcus was a man	$\text{man}(\text{Marcus})$
② Marcus was a Pompeian.	$\text{Pompeian}(\text{Marcus})$
③ All Pompeians were Romans.	$\forall x \text{ Pompeian}(x) \rightarrow \text{Roman}(x)$
④ Caesar was a Ruler	$\text{ruler}(\text{Caesar})$
⑤ All Romans were either loyal to Caesar or hated him.	$\forall x \text{ Roman}(x) \rightarrow \text{loyal to}(x, \text{Caesar}) \vee \text{hated}(x, \text{Caesar})$
⑥ Everyone is loyal to someone.	$\forall x \exists y \text{ loyal to}(x, y)$
⑦ People only try to assassinate rulers they are not loyal to.	$\forall x \forall y \text{ People}(x) \wedge \text{Ruler}(y) \wedge \text{TA}(x, y) \rightarrow \sim \text{Loyal}(x, y)$
⑧ Marcus tried to assassinate Caesar	$\text{TA}(\text{Marcus}, \text{Caesar})$
⑨ All men are people	$\forall x \text{ Man}(x) \rightarrow \text{People}(x)$

Q.1 Was Marcus loyal to Caesar?

Q.2 Was Marcus hate Caesar?

Ans 1 $\text{man}(\text{Marcus}) \xrightarrow{(9)} \text{People}(\text{Marcus}) \xrightarrow{(8)} \text{People}(\text{Marcus}) \wedge \text{TA}(\text{Marcus}, \text{Caesar}) \xrightarrow{(7)} \text{People}(\text{Marcus}) \wedge \text{ruler}(\text{Caesar}) \wedge \text{TA}(\text{Marcus}, \text{Caesar}) \xrightarrow{(3)} \sim \text{loyal}(\text{Marcus}, \text{Caesar}) \xrightarrow{(10)} \text{False}$

Ans 2 $\text{Pompeian}(\text{Marcus}) \xrightarrow{(2)} \text{Roman}(\text{Marcus}) \xrightarrow{(10)} \text{Roman}(\text{Marcus}) \xrightarrow{(5)} \sim \text{loyal}(\text{Marcus}, \text{Caesar}) \rightarrow \text{hated}(\text{Marcus}, \text{Caesar})$

Q. 2) ① Marcus was a man	$\text{man}(\text{Marcus})$
② Marcus was a Pompeian	$\text{Pompeian}(\text{Marcus})$
③ Marcus was born in 40 A.D	$\text{born}(\text{Marcus}, 40 \text{ A.D.})$
④ All men are mortal	$\forall x \text{ Man}(x) \rightarrow \text{Mortal}(x)$
⑤ All Pompeians died when the volcano erupted in 79 A.D	$\forall x \text{ Pompeian}(x) \rightarrow \text{died}(x, 79) \quad (S.i)$ $\text{erupt}(\text{volcano}, 79) \quad (S.ii)$
⑥ No mortal lives longer than 150 yrs.	$\forall x \forall t_1 \forall t_2 \text{ mortal}(x) \wedge \text{born}(x, t_1) \wedge \text{gt}(t_2 - t_1, 150) \rightarrow \text{dead}(x, t_2)$
⑦ It is now 1991	$\text{now} = 1991$
⑧ A live means not dead.	$\forall x \forall t (\text{alive}(x, t) \rightarrow \neg \text{dead}(x, t) \wedge (\text{dead}(x, t) \rightarrow \neg \text{alive}(x, t)))$
⑨ If someone dies, then he is dead at all later times.	$\forall x \forall t_1 \forall t_2 \text{ died}(x, t_1) \wedge \text{gt}(t_2, t_1) \rightarrow \text{dead}(x, t_2)$

Q. Was Marcus dead?

Ans) $\text{gt}(1991, 79) \xrightarrow{(7)} \text{gt}(\text{now}, 79) \xrightarrow{(2)} \text{Pompeian}(\text{Marcus}) \wedge \text{gt}(\text{now}, 79)$
 $\xrightarrow{(S.i)} \text{died}(\text{Marcus}, 79) \wedge \text{gt}(\text{now}, 79) \xrightarrow{(9)} \text{dead}(\text{Marcus}, \text{now})$

CNF form

- ① $\forall (x) \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 ② $\text{food}(\text{apples}) \wedge \text{food}(\text{vegetables})$
 ③ $\forall x \forall y \neg \text{eat}(x, y) \vee \text{killed}(y) \vee \text{food}(y)$
 ④ $\text{eats}(\text{Anil}, \text{peanut}) \wedge \text{alive}(\text{Anil})$
 ⑤ $\forall (x) \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
 ⑥ $\text{like}(\text{John}, \text{peanuts})$

- ⑦ $\forall x \neg \text{killed}(x) \vee \text{alive}(x)$
 ⑧ $\forall x \text{ alive}(x) \rightarrow \neg \text{killed}(x)$

Removing $\forall x$

- ① $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 ② $\text{food}(\text{apples}) \wedge \text{food}(\text{vegetables})$
 ③ $\neg \text{eat}(x, y)$

Steps to Convert Logical Statement to CNF

Steps to Convert Logical Statement into CNF

Conjunctive normal form states that a formula is in CNF, if it is a conjunctive of one or more ^{Has one} clauses, where each clause is a disjunction of literals.

Eg: $(\neg A \vee \neg B \vee C) \wedge (\neg A \vee B \vee \neg C)$

Steps

→ Replace $a \rightarrow b$ with $\neg a \vee b$ & $(a \leftrightarrow b) \equiv (a \rightarrow b) \wedge (b \rightarrow a)$

→ Push negation inside. Eg.

① $\neg(\forall x p) \equiv \exists x \neg p$

② $\neg(a \vee b) = \neg a \wedge \neg b$

③ $\neg(\exists x p) \equiv \forall x \neg p$

④ $\neg(a \wedge b) = \neg a \vee \neg b$

→ Each quantifier should use diff var.

Eg: $\exists x p(x) \wedge \forall x q(x)$ X ~~⑤~~ $\exists x p(x) \wedge \forall y q(y)$ ✓

→ Each existential quant is ~~also~~ transformed into Skolem constant or func.

Eg ① $\exists x \text{Rich}(x)$: becomes $\text{Rich}(c_1)$, c_1 = Skolem constant.

② Everyone has a heart: $\forall x \text{Person}(x) \rightarrow \exists y (\text{Heart}(y) \wedge \text{Has}(x, y))$
becomes: $\forall x \text{Person}(x) \rightarrow \text{Heart}(H(x)) \wedge \text{Has}(x, H(x))$
 H = Skolem func/const.

→ Each universal quant is ~~replaced~~ removed.

Eg: $\forall x \text{Person}(x)$ becomes $\text{Person}(x)$

→ Distribute $\wedge$ over $\vee$

Eg: $(a \wedge b) \vee c = (a \vee c) \wedge (b \vee c)$

Eg: $A \rightarrow (B \leftrightarrow C)$

$\equiv \neg A \vee (B \leftrightarrow C) \equiv \neg A \vee ((B \rightarrow C) \wedge (C \rightarrow B))$
 $\equiv (\neg A \vee (\neg B \vee C)) \wedge (\neg A \vee (\neg C \vee B)) \equiv (\neg A \vee \neg B \vee C) \wedge (\neg A \vee \neg C \vee B)$

Example: FOL to CNF

$$\forall x [\forall y \text{Animal}(y) \rightarrow \text{Loves}(x, y)] \rightarrow [\exists y \text{Loves}(y, x)]$$

Answers

Step 1: Eliminate bidirectional & implications

$$= \forall x \neg [\forall y \neg \text{Animal}(y) \vee \text{Loves}(x, y)] \vee [\exists y \text{Loves}(y, x)]$$

Step 2: Move negation inwards.

$$= \forall x [\exists y \text{Animal}(y) \wedge \neg \text{Loves}(x, y)] \vee [\exists y \text{Loves}(y, x)]$$

Step 3: Standardize variables

$$= \forall x [\exists y \text{Animal}(y) \wedge \neg \text{Loves}(x, y)] \vee [\exists z \text{Loves}(z, x)]$$

Step 4: Skolemize

$$= \forall x [\text{Animal}(F(y)) \wedge \neg \text{Loves}(x, F(y))] \vee [\text{Loves}(G(z), x)]$$

Step 5: Remove universal quantifiers

$$= [\text{Animal}(F(y)) \wedge \neg \text{Loves}(x, F(y))] \vee [\text{Loves}(G(z), x)]$$

Step 6: Distribute

$$= [\text{Animal}(F(y)) \vee \text{Loves}(G(z), x)] \wedge [\neg \text{Loves}(x, F(y)) \vee \text{Loves}(G(z), x)]$$

Resolution

Resolution $\rightarrow$

- $\rightarrow$ Is a theorem proving technique that proceeds by building proofs by contradiction
- $\rightarrow$ Used when various statements are given & we need to prove a conclusion from them
- $\rightarrow$ Is a single inference rule which can efficiently operate on the CNF

Steps in Resolution

- $\rightarrow$ Conversion of facts into FOPL
- $\rightarrow$ " " FOPL to CNF (also remove $\forall$ & $\exists$)
- $\rightarrow$ Negate the statements needed to prove (proof by contradiction)
- $\rightarrow$ Draw Resolution graph (unification)

Q.1) ① John likes all kind of food

② Apple & vegetable are food

③ Anything Anyone eats & not killed is food

④ Anil eats peanuts & still alive.

⑤ Harry eats everything that Anil eats.

⑥ John likes peanuts.

Step 1: Facts $\rightarrow$ FOPL

$\forall x \text{ food}(x) \rightarrow \text{likes}(\text{John}, x)$

$\text{food}(\text{apple}) \wedge \text{food}(\text{vegetable})$

$\forall x \forall y \text{ eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$

$\text{eats}(\text{Anil}, \text{peanut}) \wedge \text{alive}(\text{Anil})$

$\forall x \text{ eat}(\text{Anil}, x) \rightarrow \text{eat}(\text{Harry}, x)$

$\text{likes}(\text{John}, \text{peanut})$

We can also say that: $\forall x \neg \text{killed}(x) \rightarrow \text{alive}(x)$

$\forall x \text{ alive}(x) \rightarrow \neg \text{killed}(x)$

Assume: $\text{likes}(\text{John}, \text{peanut})$ needs to be proved.

Step 2: FOPL to CNF

① $\neg \forall x \neg \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$

② $\text{food}(\text{apples}) \wedge \text{food}(\text{vegetables})$

③ $\forall x \forall y \text{ eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$

④ $\text{eats}(\text{Anil}, \text{peanut}) \wedge \text{alive}(\text{Anil})$

⑤ $\forall x \neg \text{eat}(\text{Anil}, x) \vee \text{eat}(\text{Harry}, x)$

⑥ $\text{likes}(\text{John}, \text{peanut})$

⑦ $\forall x \text{ killed}(x) \rightarrow \text{alive}(x)$

⑧ $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$

Step 3

Renaming variables

$\forall y \forall z \text{ eats}(y, z) \vee$
 $\text{killed}(y) \vee \text{food}(z)$

$\forall w \neg \text{eat}(\text{Anil}, w) \vee$
 $\text{eat}(\text{Harry}, w)$

$\forall g \text{ killed}(g) \vee \text{alive}(g)$

$\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$

Step 5: Eliminate Quantifiers

① $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$

② $\text{food}(\text{apples})$

③ $\text{food}(\text{vegetables})$

④ $\neg \text{eat}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$

⑤ $\text{eats}(\text{Anil}, \text{peanut})$

⑥ $\text{alive}(\text{Anil})$

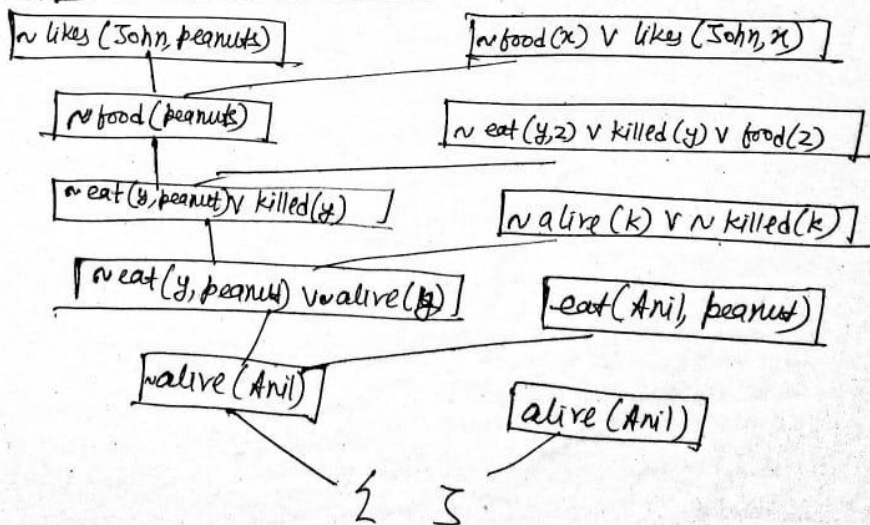
⑦ $\neg \text{eat}(\text{Anil}, w) \vee \text{eat}(\text{Harry}, w)$

⑧ $\text{likes}(\text{John}, \text{peanut})$

⑨ $\text{killed}(g) \vee \text{alive}(g)$

⑩ $\neg \text{alive}(k) \vee \neg \text{killed}(k)$

Step 6: Draw Resolution Tree



Empty set means. $\neg \text{likes}(\text{John}, \text{peanuts})$ is false.
 $\therefore \text{likes}(\text{John}, \text{peanuts})$ is True.

Example

The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles. All of its missiles were sold to it by Colonel West, who is American. An enemy of American counts as hostile.

Prove By Resolution: West is criminal.

Answer

Step 1: Facts to FOL

Rule 1: It is crime for an American to sell weapons to hostile nations

$$\forall x, y, z [American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z)] \Rightarrow Criminal(x)$$

Rule 2: Nono has some missiles.

$$\exists x [Owns(Nono, x) \wedge Missiles(x)]$$

Rule 3: Nono has missiles sold to West.

$$\forall x Missiles(x) \wedge Owns(Nono, x) \rightarrow Sells(West, x, Nono)$$

Rule 4: $\forall x Enemy(x, America) \rightarrow Hostile(x)$

Rule 5: American(West)

Rule 6: Enemy(Nono, America)

Rule 7: $\forall x Missile(x) \rightarrow Weapon(x)$

Step 1: $\forall x, y, z [American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z)] \rightarrow Criminal(x)$

2) Owns(Nono, M1)

3) Missile(M1)

4) $\forall x [Missile(x) \wedge Owns(Nono, x) \rightarrow Sells(West, x, Nono)]$

5) $\forall x Enemy(x, America) \rightarrow Hostile(x)$

6) American(West)

7) Enemy(Nono, America)

8) $\forall x Missile(x) \rightarrow Weapon(x)$

Step 2: FOL to CNF

i) Eliminate $\rightarrow$ & $\leftrightarrow$ & move negs inside

1) $\forall x, y, z [American(x) \vee \neg Weapon(y) \vee \neg Sells(x, y, z) \vee \neg Hostile(z) \vee Criminal(x)]$

2) Owns(Nono, M1)

3) Missile(M1)

4) $\forall x [\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)]$

5) $\forall x \neg Enemy(x, America) \vee Hostile(x)$

6) American(West)

7) Enemy(Nono, America)

8) $\forall x \neg Missile(x) \vee Weapon(x)$

ii) No standardize var & Skolemize nodes
Remove universal quant

1) $\neg American(x) \vee \neg Weapon(y) \vee \neg Sells(x, y, z) \vee \neg Hostile(z) \vee Criminal(x)$

2) Owns(Nono, M1)

3) Missile(M1)

4) $\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)$

5) $\neg Enemy(x, America) \vee Hostile(x)$

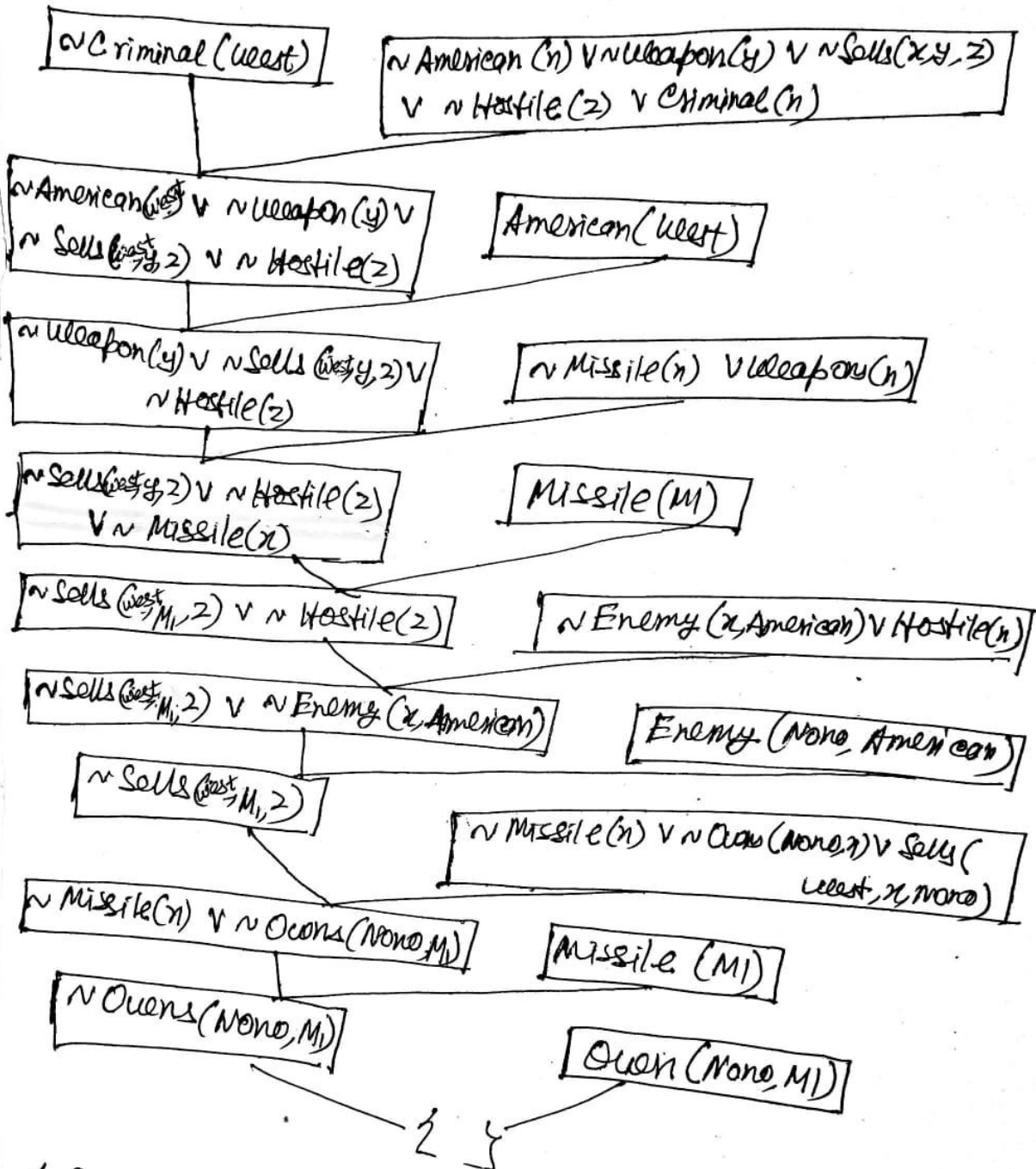
6) American(West)

7) Enemy(Nono, America)

8) $\neg Missile(x) \vee Weapon(x)$

Step 3: Resolution Tree.

To prove: $\text{West is Criminal} \equiv \text{Criminal}(\text{West})$



$\bot$ means, $\neg \text{Criminal}(\text{West})$ is false by contradiction.
 $\therefore \text{Criminal}(\text{West})$ is true.

Bayesian Networks

A **Bayesian Network** (also known as a **Bayes Network** or **Belief Network**) is a **probabilistic graphical model** that represents a set of variables and their conditional dependencies through a directed acyclic graph (DAG). Due to its feature of joint probability, the probability in Bayesian Belief Network is derived, based on a condition — $P(\text{attribute/parent})$ i.e probability of an attribute, true over parent attribute.

It is used for reasoning under uncertainty, where the relationships between variables are expressed in terms of probability theory.

Components of a Bayesian Network:

1. **Nodes (Variables):** Each node in the network represents a random variable, which can be discrete or continuous. These variables can represent various phenomena or events (e.g., weather, diseases, sensor readings).
2. **Edges (Dependencies):** The directed edges between nodes represent **conditional dependencies**. If there is a directed edge from node A to node B, it implies that **B is conditionally dependent** on A.
3. **Conditional Probability Distribution (CPD):** Each node has a **conditional probability distribution** that defines the probability of the node given its parent nodes. For example, the probability of an event occurring given certain other events.
4. **Joint Probability Distribution:** A Bayesian network encodes the joint probability distribution of the set of variables by factoring it into a product of conditional probabilities.

How Bayesian Networks Work:

1. **Inference:** Given some observed data (evidence), Bayesian networks can be used to **infer** the probabilities of unknown variables. This allows for reasoning about uncertain events based on known information.
2. **Learning:** Bayesian networks can also be **learned** from data, either **parameter learning** (estimating the CPD values) or **structure learning** (determining the network's structure).

Applications of Bayesian Networks:

1. **Medical Diagnosis:** Bayesian networks are used to model the relationships between symptoms, diseases, and medical tests. They allow medical experts to reason about the likelihood of a disease given the presence of symptoms and test results.
2. **Decision Support Systems:** In uncertain environments, Bayesian networks help in making decisions based on incomplete information by calculating the probabilities of various outcomes.
3. **Risk Analysis:** Bayesian networks can model risks and uncertainties in systems such as engineering, finance, and project management.
4. **Natural Language Processing (NLP):** Used in machine translation, speech recognition, and text prediction by modelling the probabilistic dependencies between words or phrases.

Advantages of Bayesian Networks:

1. **Handling Uncertainty:** Bayesian networks provide a formalism for reasoning with uncertain information.
2. **Modular:** They are flexible and can be easily extended with new variables or dependencies.
3. **Interpretability:** The graphical representation makes it easy to interpret the relationships between variables.

Limitations of Bayesian Networks:

1. **Computational Complexity:** Exact inference in large networks can be computationally expensive due to the large number of possible combinations of variables.
2. **Data Availability:** Learning the structure or parameters of a Bayesian network requires significant data, especially when dealing with complex networks.

Example of a Simple Bayesian Network: Imagine a simple network with the following variables:

- **Rain:** Whether it's raining (True/False).
- **Traffic:** Whether there is traffic (True/False).
- **Accident:** Whether there is an accident (True/False).

The network might look like this: ***Rain* → *Traffic* → *Accident***

Where:

- The probability of traffic is dependent on whether it's raining.
- The probability of an accident is dependent on both traffic and rain.

The CPDs might look like:

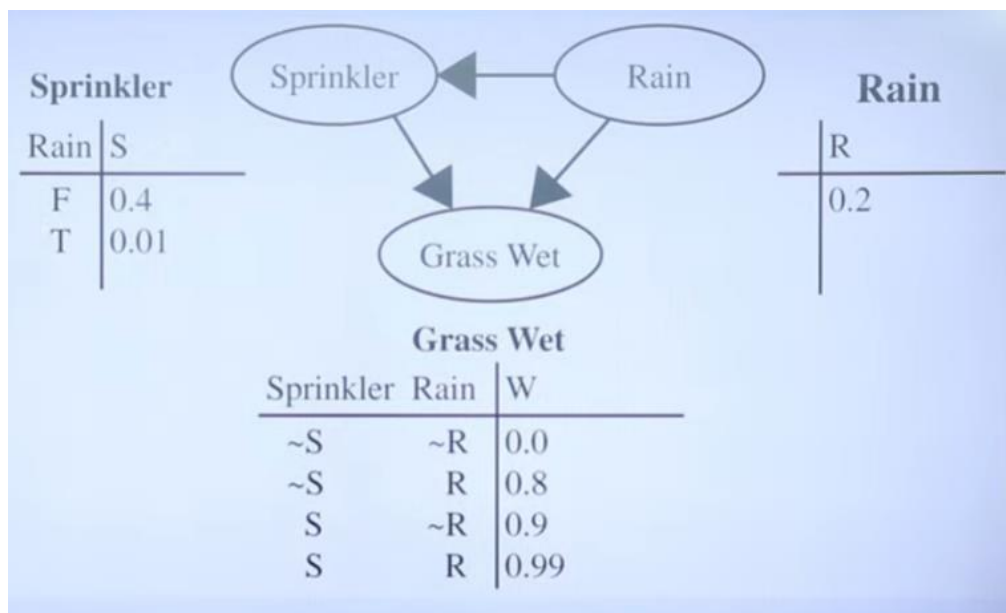
- **P(Traffic | Rain):** The probability of traffic given that it's raining.
- **P(Accident | Traffic, Rain):** The probability of an accident given the traffic and rain.

Bayesian Network Representation:

- **Rain** and **Traffic** have independent probabilities.
- **Traffic** depends on **Rain**, and **Accident** depends on both **Traffic** and **Rain**.

Question 1: Find the probability

- i) **when the grass is wet and there is sprinkler and rain.**
- ii) **When the grass is wet but no sprinkler and no rain**



Answer

R	$\sim R$
0.2	0.8

R	S	$\sim S$
F	0.4	0.6
T	0.01	0.99

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

S	R	W	$\sim W$
$\sim S$	$\sim R$	0	1
$\sim S$	R	0.8	0.2
S	$\sim R$	0.9	0.1
S	R	0.99	0.01

i) $P(W \cap S \cap R) = ?$

$$= P(W|S, R) \times P(S|R) \times P(R)$$

$$= 0.99 \times 0.01 \times 0.2$$

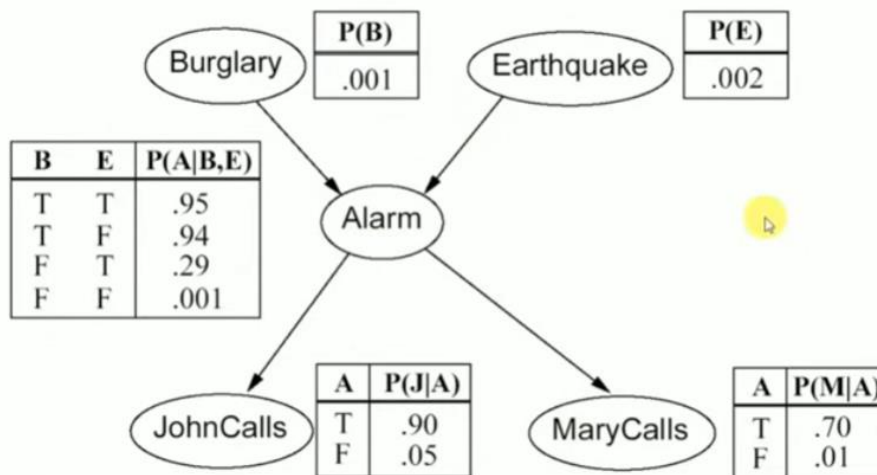
$$= 0.00198$$

ii) $P(W \cap \sim S \cap \sim R) = P(W|\sim S, \sim R) \times P(\sim S|\sim R) \times P(\sim R)$

$$= 0 \times 0.6 \times 0.8 = 0$$

Question 2: Find the probability that:

- The alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry called.
- John called.
- There is a burglary given that John and Merry calls.



- You have a new burglar alarm installed at home.
- It is fairly reliable at detecting burglary, but also sometimes responds to minor earthquakes.
- You have two neighbors, John and Merry, who promised to call you at work when they hear the alarm.
- John always calls when he hears the alarm, but sometimes confuses telephone ringing with the alarm and calls too.
- Merry likes loud music and sometimes misses the alarm.
- Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

Answer

$$\begin{aligned} \text{i) } P(A \cap \neg B \cap \neg E \cap J \cap M) &= P(A | \neg B, \neg E) \times P(\neg B) \times P(\neg E) \times P(J | A) \times P(M) \\ &= 0.001 \times 0.999 \times 0.992 \times 0.9 \times 0.7 \\ &= 0.00062 \end{aligned}$$

ii) John can call in 2 cases — he heard alarm
— he not heard alarm but thought telephone as alarm

$$= P(J | A) P(A) + P(J | \neg A) P(\neg A)$$

$$\begin{array}{l|l} \text{Now, } P(A) = 0.95 \times 0.001 \times 0.002 & P(\neg A) = 0.05 \times 0.001 \times 0.002 \\ + 0.94 \times 0.001 \times 0.998 & + 0.06 \times 0.001 \times 0.998 \\ + 0.29 \times 0.999 \times 0.002 & + 0.71 \times 0.999 \times 0.002 \\ + 0.001 \times 0.999 \times 0.998 & + 0.999 \times 0.999 \times 0.998 \\ \hline = 0.00252 & = 0.99748 \end{array}$$

$$\begin{aligned} &= 0.90 \times 0.00252 + 0.05 \times 0.99748 \\ &= 0.052143 \end{aligned}$$

Dempster-Shafer Theory (DST)

The **Dempster-Shafer Theory (DST)**, also known as the **Theory of Evidence**, is a mathematical framework for modelling and reasoning under uncertainty. It provides a method for combining all evidence from different sources and arriving at a degree of belief (also called "belief function") about an event or hypothesis.

Unlike traditional probability theory, DST allows for **partial belief** and **ignorance** when the information available is uncertain or incomplete.

This theory was released because of the following reason:

- Bayesian theory is only concerned about single evidence.
- Bayesian probability cannot describe ignorance.

Key Concepts in Dempster-Shafer Theory:

1. **Frame of Discernment (Θ):** The set of all possible outcomes (or hypotheses) in each situation.
Example: For a weather prediction system, the frame of discernment might be **{Rain, No Rain}**.
2. **Basic Probability Assignment (BPA):** A function that assigns probabilities to subsets of the frame of discernment. This function reflects the degree of belief in each subset based on available evidence.
 - a. Denoted as $m(A)$, where A is a subset of the frame of discernment.
 - b. Conditions:
 - i. $m(\emptyset) = 0$, meaning the empty set has zero probability.
 - ii. The sum of all BPAs must be: $\sum_{A \subseteq \Theta} m(A) = 1$.

3. **Belief Function (Bel):** The belief function gives the total amount of belief that supports a hypothesis based on the available evidence.

$$\text{Bel}(A) = \sum_{B \subseteq A} m(B), \text{ where } B \text{ is a subset of } A.$$

4. **Plausibility Function (Pl):** –The plausibility function represents how likely a hypothesis is, considering all evidence and excluding the possibility of disbelief.

$$\text{Pl}(A) = 1 - \text{Bel}(\neg A), \text{ where } \text{Bel}(\neg A) \text{ is the belief in the complement of } A.$$

5. **Disbelief Function (Disbelief or Dis):** The disbelief function represents the degree to which a hypothesis is not supported by the evidence. It is complementary to belief.

$$\text{Dis}(A) = 1 - \text{Pl}(A)$$

6. **Combination Rule (Dempster's Rule of Combination):** Dempster's Rule is a method for combining multiple pieces of evidence to derive a more complete belief.

- a. The combination rule combines two belief functions, **m1** and **m2**, into a new belief function **m**:

- $$m(A) = \sum_{B \cap C = A} \frac{m_1(B) \cdot m_2(C)}{1 - K}$$
 - **K** is a normalization factor to handle conflicts:

$$K = \sum_{B \cap C = \emptyset} m_1(B) \cdot m_2(C)$$
 - This rule is used when there is overlap between sources of evidence and enables the merging of independent sources of information.

Example: Scenario: Diagnosing a Disease (Flu vs. COVID-19)

You are trying to determine whether a patient has **Flu** or **COVID-19**, based on two sources of evidence:

1. **Symptom Evidence (Source 1):** Fever, cough, and fatigue.
2. **Test Results Evidence (Source 2):** Results from a rapid test.

1. Frame of Discernment (Θ)

The possible outcomes or hypotheses: $\Theta = \{\text{Flu}, \text{COVID-19}, \text{Neither}\}$

2. Basic Probability Assignments (BPA)

Each source provides the following belief values:

Source 1: Symptom Evidence

- $m_1(\text{Flu}) = 0.5$ (50% belief symptoms are due to Flu).
- $m_1(\text{COVID-19}) = 0.3$ (30% belief symptoms are due to COVID-19).
- $m_1(\text{Neither}) = 0.2$ (20% belief symptoms are unrelated to either).

Source 2: Test Results Evidence

- $m_2(\text{Flu}) = 0.4$ (40% belief test indicates Flu).
- $m_2(\text{COVID-19}) = 0.5$ (50% belief test indicates COVID-19).
- $m_2(\text{Neither}) = 0.1$ (10% belief test is inconclusive).

3. Combining Evidence (Dempster's Rule of Combination)

We combine m_1 and m_2 using Dempster's rule:

Step 1: Calculate Combined Beliefs

- $$m(\text{Flu}) = \frac{\sum \text{belief combinations supporting Flu}}{1-K}$$
$$= \frac{[m_1(\text{Flu}) \times m_2(\text{Flu}) + m_1(\text{Flu}) \times m_2(\text{Neither}) + m_1(\text{Neither}) \times m_2(\text{Flu})]}{1-K}$$
$$= \frac{[0.5 \times 0.4 + 0.5 \times 0.1 + 0.2 \times 0.4]}{1-K}$$
$$= \frac{0.2 + 0.05 + 0.08}{1-K} = \frac{0.33}{1-K}$$
- Similarly, calculate $m(\text{COVID-19})$ and $m(\text{Neither})$.

Step 2: Resolve Conflicts (Normalization)

- Calculate conflict factor $K =$
$$\sum \text{belief combinations with no overlap:}$$
$$K = m_1(\text{Flu}) \times m_2(\text{COVID-19}) + m_1(\text{COVID-19}) \times m_2(\text{Flu}) = 0.5 \times 0.5 + 0.3 \times 0.4 = 0.25 + 0.12 = 0.37.$$
- Normalize results by dividing $1 - K$.

4. Final Belief Distribution

After combining and normalizing:

- $m(\text{Flu}) = 0.40$ (40% belief patient has Flu).
- $m(\text{COVID-19}) = 0.50$ (50% belief patient has COVID-19).
- $m(\text{Neither}) = 0.10$ (10% belief symptoms are unrelated).

Interpretation: The combined evidence suggests a **higher belief (50%) that the patient has COVID-19**, though there is still a 40% belief that it could be Flu. Further tests or evidence may help reduce uncertainty.

Applications of Dempster-Shafer Theory:

1. **Decision Support Systems:** DST is widely used in fields where information is uncertain or incomplete, such as medical diagnosis, expert systems, and risk analysis.
2. **Sensor Fusion:** DST is effective in combining data from different sensors with varying levels of reliability, where conflicts in the data can arise.
3. **Artificial Intelligence:** DST helps AI systems reason with uncertain knowledge, such as autonomous vehicles using DST to combine sensor data for decision-making.
4. **Robotics:** In robotics, DST helps robots to process and combine sensory inputs to make decisions under uncertainty.

Advantages of Dempster-Shafer Theory:

1. **Handling Uncertainty:** DST can deal with both uncertainty and **ignorance** (lack of information), which is a major advantage over traditional probability theory.
2. **Flexibility:** DST allows for more **flexible reasoning** by representing partial beliefs and conflicts.
3. **Combining Evidence:** DST provides a formal way to combine conflicting evidence from different sources, making it useful in multi-source decision-making.

Limitations of Dempster-Shafer Theory:

1. **Computational Complexity:** Combining large amounts of evidence can lead to computational challenges.

2. **Conflict Handling:** The normalization factor KKK can be very small when there is significant conflict between sources, leading to loss of information.
3. **No clear rule for handling "negative" evidence or strongly conflicting data.**

Fuzzy Sets

A **fuzzy set** is a class of objects with **unprecise boundaries**. Unlike classical sets (where an element either belongs to the set or does not), fuzzy sets allow for **partial membership**. Each element in a fuzzy set has a degree of membership, which is a value between 0 and 1. **Key Concepts:**

- **Membership Function:** A function that assigns a degree of membership (between 0 and 1) to each element of the universe of discourse.
- **Universal Set:** The set of all possible elements in the domain being considered.
- **Degree of Membership (μ_A):** A value between 0 and 1 indicating how strongly an element belongs to the fuzzy set. For example, in a fuzzy set "tall people," a person who is 6 feet tall may have a degree of membership of 0.8.

Example:

Consider the fuzzy set "Tall People" in the domain of human height.

- **Universe of Discourse:** The set of all possible human heights: $\{4.5, 5.0, 5.5, 6.0, 6.5, 7.0\}$ feet.
- **Membership Function:**
 - $\mu_{\text{Tall}}(5.5) = 0.5$
 - $\mu_{\text{Tall}}(6.0) = 0.8$
 - $\mu_{\text{Tall}}(7.0) = 1.0$

In this case, 6 feet tall individuals are considered fairly tall with a membership value of 0.8, whereas 5.5 feet tall individuals only partially belong to the set with a membership value of 0.5.

Fuzzy Logic

Fuzzy Logic is a mathematical system that allows reasoning about **imprecise or vague concepts** by using fuzzy sets. It extends classical (binary) logic by allowing truth values to range between **completely true (1) and completely false (0)**. Instead of using only 0 or 1, fuzzy logic uses a range of values between 0 and 1 to represent the **degree of truth**. **Basic Concepts in Fuzzy Logic:**

- **Fuzzy Variables:** Variables that can take on a range of values, not just discrete ones. For instance, "temperature" can range from "cold" to "hot," with various degrees of "warm."
- **Fuzzy Rules:** The basis of fuzzy logic is a set of IF-THEN rules that describe how to map fuzzy inputs to fuzzy outputs. **Example:** IF the temperature is "high," THEN the fan speed should be "fast."
- **Fuzzy Inference System (FIS):** The framework that maps inputs to outputs based on fuzzy rules. It processes fuzzy input values and applies fuzzy logic operations to produce a fuzzy output.

Key Operations in Fuzzy Logic:

- **Fuzzy AND (Intersection):** The minimum of two membership values.
 - $\mu_{A \text{ AND } B}(x) = \min(\mu_A(x), \mu_B(x))$
- **Fuzzy OR (Union):** The maximum of two membership values.
 - $\mu_{A \text{ OR } B}(x) = \max(\mu_A(x), \mu_B(x))$
- **Fuzzy NOT (Negation):** The complement of a membership value.
 - $\mu_{\text{NOT } A}(x) = 1 - \mu_A(x)$

Example:

Let's say we want to control the speed of a fan based on the temperature. We define fuzzy sets for "Temperature" and "Fan Speed":

- **Temperature:** Cold, Warm, Hot.
- **Fan Speed:** Low, Medium, High.

The fuzzy rules could look like:

- IF Temperature is "Hot" THEN Fan Speed is "High".
- IF Temperature is "Warm" THEN Fan Speed is "Medium".
- IF Temperature is "Cold" THEN Fan Speed is "Low".

If the temperature is 75°F, it could have a membership value of 0.5 in "Warm" and 0.5 in "Hot." The fuzzy inference system combines these degrees of membership and outputs a fuzzy fan speed value, which is then defuzzified (explained below) to provide a crisp output.

Fuzzy logic Operations

Example 1: 2 fuzzy sets A & B. In A degree of membership for element 2,4,6,8 are 1,0.3,0.5,0.2. In B degree of membership for element 2,4,6,8 are 0.5,0.4,0.1,1. Find union, intersection, complement and difference between these.

$$\begin{aligned} \underline{A} &= \left\{ \frac{1}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0.2}{8} \right\} \\ \underline{B} &= \left\{ \frac{0.5}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{1}{8} \right\} \end{aligned}$$

Solution:

UNION $\underline{A} \cup \underline{B} = \max\{\mu_{\underline{A}}(x), \mu_{\underline{B}}(x)\} = \left\{ \frac{1}{2} + \frac{0.4}{4} + \frac{0.5}{6} + \frac{1}{8} \right\}$	COMPLEMENT $\underline{A} = 1 - \mu_{\underline{A}}(x) = \left\{ \frac{0}{2} + \frac{0.7}{4} + \frac{0.5}{6} + \frac{0.8}{8} \right\}$
INTERSECTION $\underline{A} \cap \underline{B} = \min\{\mu_{\underline{A}}(x), \mu_{\underline{B}}(x)\} = \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.1}{6} + \frac{0.2}{8} \right\}$	$\underline{B} = 1 - \mu_{\underline{B}}(x) = \left\{ \frac{0.5}{2} + \frac{0.6}{4} + \frac{0.9}{6} + \frac{0}{8} \right\}$
DIFFERENCE $\underline{A} \setminus \underline{B} = \underline{A} \cap \overline{\underline{B}} = \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0}{8} \right\}$	
$\underline{B} \setminus \underline{A} = \underline{B} \cap \overline{\underline{A}} = \left\{ \frac{0}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{0.8}{8} \right\}$	

Example 2:

$$\tilde{B}_1 = \left\{ \frac{1}{1.0} + \frac{0.75}{1.5} + \frac{0.3}{2.0} + \frac{0.15}{2.5} + \frac{0}{3.0} \right\}$$

$$\tilde{B}_2 = \left\{ \frac{1}{1.0} + \frac{0.6}{1.5} + \frac{0.2}{2.0} + \frac{0.1}{2.5} + \frac{0}{3.0} \right\}$$

Solution:

$$\bullet \tilde{B}_1 \cup \tilde{B}_2 = \left\{ \frac{1}{1.0} + \frac{0.75}{1.5} + \frac{0.3}{2.0} + \frac{0.15}{2.5} + \frac{0}{3.0} \right\}$$

$$\bullet \overline{\tilde{B}_1} = \left\{ \frac{0}{1.0} + \frac{0.25}{1.5} + \frac{0.7}{2.0} + \frac{0.85}{2.5} + \frac{1}{3.0} \right\}$$

$$\bullet \tilde{B}_1 \cap \tilde{B}_2 = \left\{ \frac{1}{1.0} + \frac{0.6}{1.5} + \frac{0.2}{2.0} + \frac{0.1}{2.5} + \frac{0}{3.0} \right\}$$

$$\bullet \overline{\tilde{B}_2} = \left\{ \frac{0}{1.0} + \frac{0.4}{1.5} + \frac{0.8}{2.0} + \frac{0.9}{2.5} + \frac{1}{3.0} \right\}$$

$$\bullet \tilde{B}_1 | \tilde{B}_2 = \tilde{B}_1 \cap \overline{\tilde{B}_2} = \left\{ \frac{0}{1.0} + \frac{0.4}{1.5} + \frac{0.3}{2.0} + \frac{0.15}{2.5} + \frac{0}{3.0} \right\}$$

$$\bullet \overline{\tilde{B}_1 \cup \tilde{B}_2} = \left\{ \frac{0}{1.0} + \frac{0.25}{1.5} + \frac{0.7}{2.0} + \frac{0.85}{2.5} + \frac{1}{3.0} \right\}$$

Example 3: It is necessary to compare two sensors based upon their detection levels and gain settings. The table of gain settings and sensor detection levels with a standard item being monitored providing typical membership values to represent the detection levels for each sensor is given in the table.

Now given the universe of discourse $X = \{0, 10, 20, 30, 40, 50\}$

and the membership functions for the two sensors in discrete form as

$$\tilde{D}_1 = \left\{ \frac{0}{0} + \frac{0.2}{10} + \frac{0.35}{20} + \frac{0.65}{30} + \frac{0.85}{40} + \frac{1}{50} \right\}$$

$$\tilde{D}_2 = \left\{ \frac{0}{0} + \frac{0.35}{10} + \frac{0.25}{20} + \frac{0.8}{30} + \frac{0.95}{40} + \frac{1}{50} \right\}$$

Gain	Detection Level Sensor 1	Detection Level Sensor 2
0	0	0
10	0.2	0.35
20	0.35	0.25
30	0.65	0.8
40	0.85	0.95
50	1	1

Solution:

$$(a) \mu_{\tilde{D}_1 \cup \tilde{D}_2}(x) = \left\{ \frac{0}{0} + \frac{0.35}{10} + \frac{0.35}{20} + \frac{0.8}{30} + \frac{0.95}{40} + \frac{1}{50} \right\}$$

$$(d) \mu_{\tilde{D}_1 \cup \overline{\tilde{D}_1}}(x) = \max\{\mu_{\tilde{D}_1}(x), \mu_{\overline{\tilde{D}_1}}(x)\}$$

$$= \left\{ \frac{1}{0} + \frac{0.8}{10} + \frac{0.65}{20} + \frac{0.65}{30} + \frac{0.85}{40} + \frac{1}{50} \right\}$$

$$(b) \mu_{\tilde{D}_1 \cap \tilde{D}_2}(x) = \left\{ \frac{0}{0} + \frac{0.2}{10} + \frac{0.25}{20} + \frac{0.65}{30} + \frac{0.85}{40} + \frac{1}{50} \right\}$$

$$(c) \mu_{\overline{\tilde{D}_1}}(x) = \left\{ \frac{1}{0} + \frac{0.8}{10} + \frac{0.65}{20} + \frac{0.35}{30} + \frac{0.15}{40} + \frac{0}{50} \right\}$$

$$(e) \mu_{\tilde{D}_1 \cap \overline{\tilde{D}_1}}(x)$$

$$= \left\{ \frac{0}{0} + \frac{0.2}{10} + \frac{0.35}{20} + \frac{0.35}{30} + \frac{0.15}{40} + \frac{0}{50} \right\}$$

$$\mu_{\overline{\tilde{D}_2}}(x) = \left\{ \frac{1}{0} + \frac{0.65}{10} + \frac{0.75}{20} + \frac{0.2}{30} + \frac{0.05}{40} + \frac{0}{50} \right\}$$

Algebraic sum & product, bounded sum & difference

Example 1: Find the algebraic sum, algebraic product, bounded sum and bounded difference of the given fuzzy sets

$$\tilde{A} = \left\{ \frac{0.2}{1} + \frac{0.3}{2} + \frac{0.4}{3} + \frac{0.5}{4} \right\}$$

$$\tilde{B} = \left\{ \frac{0.1}{1} + \frac{0.2}{2} + \frac{0.2}{3} + \frac{1}{4} \right\}$$

Solution

$$(a) \text{ Algebraic sum} = \mu_{\tilde{A} + \tilde{B}}(x) = [\mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x)] - [\mu_{\tilde{A}}(x) \cdot \mu_{\tilde{B}}(x)]$$

$$= \left\{ \frac{0.3}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\} - \left\{ \frac{0.02}{1} + \frac{0.06}{2} + \frac{0.08}{3} + \frac{0.5}{4} \right\} = \left\{ \frac{0.28}{1} + \frac{0.44}{2} + \frac{0.52}{3} + \frac{0}{4} \right\}$$

$$(b) \text{ Algebraic product} = \mu_{\tilde{A} \cdot \tilde{B}}(x) = \mu_{\tilde{A}}(x) \cdot \mu_{\tilde{B}}(x) = \left\{ \frac{0.02}{1} + \frac{0.06}{2} + \frac{0.08}{3} + \frac{0.5}{4} \right\}$$

$$(c) \text{ Bounded sum} = \mu_{\tilde{A} \oplus \tilde{B}}(x) = \min[1, \mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x)]$$

$$= \min \left\{ 1, \left\{ \frac{0.3}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\} \right\} = \left\{ \frac{0.3}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\}$$

$$(d) \text{ Bounded difference} = \mu_{\tilde{A} \ominus \tilde{B}}(x) = \max[0, \mu_{\tilde{A}}(x) - \mu_{\tilde{B}}(x)]$$

$$= \max \left\{ 0, \left\{ \frac{0.1}{1} + \frac{0.1}{2} + \frac{0.2}{3} + \frac{0.5}{4} \right\} \right\} = \left\{ \frac{0.1}{1} + \frac{0.1}{2} + \frac{0.2}{3} + \frac{0.5}{4} \right\}$$

Example 2: Find the algebraic sum, algebraic product, bounded sum, and bounded difference of the given fuzzy sets

$$\mu_{\tilde{T}} = \left\{ \frac{0}{0} + \frac{0.2}{1} + \frac{0.7}{2} + \frac{0.8}{3} + \frac{0.9}{4} + \frac{1}{5} \right\}$$

$$\mu_{\tilde{R}} = \left\{ \frac{0}{0} + \frac{0.1}{1} + \frac{0.3}{2} + \frac{0.2}{3} + \frac{0.4}{4} + \frac{0.5}{5} \right\}$$

Solution:

$$(a) \text{ Algebraic sum} = \left\{ \frac{0}{0} + \frac{0.3}{1} + \frac{1.0}{2} + \frac{1.0}{3} + \frac{1.3}{4} + \frac{1.5}{5} \right\} - \left\{ \frac{0}{0} + \frac{0.02}{1} + \frac{0.21}{2} + \frac{0.16}{3} + \frac{0.36}{4} + \frac{0.5}{5} \right\}$$

$$= \left\{ \frac{0}{0} + \frac{0.28}{1} + \frac{0.79}{2} + \frac{0.84}{3} + \frac{0.94}{4} + \frac{1}{5} \right\}$$

$$(b) \text{ Algebraic product} = \left\{ \frac{0}{0} + \frac{0.02}{1} + \frac{0.21}{2} + \frac{0.16}{3} + \frac{0.36}{4} + \frac{0.5}{5} \right\}$$

$$(c) \text{ Bounded sum} = \min \left\{ 1, \left\{ \frac{0}{0} + \frac{0.3}{1} + \frac{1.0}{2} + \frac{1.0}{3} + \frac{1.3}{4} + \frac{1.5}{5} \right\} \right\} = \left\{ \frac{0}{0} + \frac{0.3}{1} + \frac{1.0}{2} + \frac{1.0}{3} + \frac{1.0}{4} + \frac{1.0}{5} \right\}$$

$$(d) \text{ Bounded difference} = \max \left\{ 0, \left\{ \frac{0}{0} + \frac{0.1}{1} + \frac{0.4}{2} + \frac{0.6}{3} + \frac{0.5}{4} + \frac{0.5}{5} \right\} \right\}$$

Cartesian Product

Example: Find the cartesian product of A and B

Solution:

$$\underline{A} = \left\{ \frac{0.3}{x_1} + \frac{0.7}{x_2} + \frac{1}{x_3} \right\}$$

$$\underline{B} = \left\{ \frac{0.4}{y_1} + \frac{0.9}{y_2} \right\}$$

$$\begin{aligned} \mu_{\underline{R}}(x_1, y_1) &= \min[\mu_{\underline{A}}(x_1), \mu_{\underline{B}}(y_1)] \\ &= \min(0.3, 0.4) = 0.3 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{R}}(x_2, y_2) &= \min[\mu_{\underline{A}}(x_2), \mu_{\underline{B}}(y_2)] \\ &= \min(0.7, 0.9) = 0.7 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{R}}(x_1, y_2) &= \min[\mu_{\underline{A}}(x_1), \mu_{\underline{B}}(y_2)] \\ &= \min(0.3, 0.9) = 0.3 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{R}}(x_3, y_1) &= \min[\mu_{\underline{A}}(x_3), \mu_{\underline{B}}(y_1)] \\ &= \min(1, 0.4) = 0.4 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{R}}(x_2, y_1) &= \min[\mu_{\underline{A}}(x_2), \mu_{\underline{B}}(y_1)] \\ &= \min(0.7, 0.4) = 0.4 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{R}}(x_3, y_2) &= \min[\mu_{\underline{A}}(x_3), \mu_{\underline{B}}(y_2)] \\ &= \min(1, 0.9) = 0.9 \end{aligned}$$

$$\underline{R} = \begin{bmatrix} 0.3 & 0.3 \\ 0.4 & 0.7 \\ 0.4 & 0.9 \end{bmatrix}$$

Max Min composition & Max product composition

Example: Find the Max-Min composition and Max- Product composition

$$\underline{R} = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.3 \\ 0.2 & 0.9 \end{bmatrix} \end{matrix} \quad \underline{S} = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.8 & 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

Solution:

Max Min Composition:

$$\begin{aligned} \mu_{\underline{T}}(x_1, z_1) &= \max\{\min[\mu_{\underline{R}}(x_1, y_1), \mu_{\underline{S}}(y_1, z_1)], \\ &\quad \min[\mu_{\underline{R}}(x_1, y_2), \mu_{\underline{S}}(y_2, z_1)]\} \\ &= \max[\min(0.6, 1), \min(0.3, 0.8)] \\ &= \max(0.6, 0.3) = 0.6 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{T}}(x_1, z_2) &= \max[\min(0.6, 0.5), \min(0.3, 0.4)] \\ &= \max(0.5, 0.3) = 0.5 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{T}}(x_1, z_3) &= \max[\min(0.6, 0.3), \min(0.3, 0.7)] \\ &= \max(0.3, 0.3) = 0.3 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{T}}(x_2, z_1) &= \max[\min(0.2, 1), \min(0.9, 0.8)] \\ &= \max(0.2, 0.8) = 0.8 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{T}}(x_2, z_2) &= \max[\min(0.2, 0.5), \min(0.9, 0.4)] \\ &= \max(0.2, 0.4) = 0.4 \end{aligned}$$

$$\begin{aligned} \mu_{\underline{T}}(x_2, z_3) &= \max[\min(0.2, 0.3), \min(0.9, 0.7)] \\ &= \max(0.2, 0.7) = 0.7 \end{aligned}$$

$$\underline{T} = \underline{R} \circ \underline{S} = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.5 & 0.3 \\ 0.8 & 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

Max Product Composition:

$$\begin{aligned}\mu_T(x_1, z_1) &= \max\{[\mu_R(x_1, y_1) \cdot \mu_S(y_1, z_1)], \\ &\quad [\mu_R(x_1, y_2) \cdot \mu_S(y_2, z_1)]\} \\ &= \max(0.6, 0.24) = 0.6\end{aligned}$$

$$\begin{aligned}\mu_T(x_1, z_2) &= \max[(0.6 \times 0.5), (0.3 \times 0.4)] \\ &= \max(0.3, 0.12) = 0.3\end{aligned}$$

$$\begin{aligned}\mu_T(x_1, z_3) &= \max[(0.6 \times 0.3), (0.3 \times 0.7)] \\ &= \max(0.18, 0.21) = 0.21\end{aligned}$$

$$\begin{aligned}\mu_T(x_2, z_1) &= \max[(0.2 \times 1), (0.9 \times 0.8)] \\ &= \max(0.2, 0.72) = 0.72\end{aligned}$$

$$\begin{aligned}\mu_T(x_2, z_2) &= \max[(0.2 \times 0.5), (0.9 \times 0.4)] \\ &= \max(0.1, 0.36) = 0.36\end{aligned}$$

$$\begin{aligned}\mu_T(x_2, z_3) &= \max[(0.2 \times 0.3), (0.9 \times 0.7)] \\ &= \max(0.06, 0.63) = 0.63\end{aligned}$$

$$\tilde{T} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.3 & 0.21 \\ 0.72 & 0.36 & 0.63 \end{bmatrix} \end{matrix}$$

Membership value for triangles using inference approach

Example: Using the inference approach, find the membership values for the triangular shapes I, R, E, IR, T for a triangle with angles 45, 55 and 80 degrees.

Solution:

Let the universe of discourse be

$$U = \{(X \ Y \ Z) : X = 80^\circ \geq Y = 55^\circ \geq Z = 45^\circ \text{ and } X + Y + Z = 80^\circ + 55^\circ + 45^\circ = 180^\circ\}$$

Membership value of isosceles triangle, I :

$$\begin{aligned}\mu_I &= 1 - \frac{1}{60^\circ} \min(X - Y, Y - Z) \\ &= 1 - \frac{1}{60^\circ} \min(80^\circ - 55^\circ, 55^\circ - 45^\circ) \\ &= 1 - \frac{1}{60^\circ} \min(25^\circ, 10^\circ) \\ &= 1 - \frac{1}{60^\circ} \times 10^\circ \\ &= 1 - 0.1667 = 0.833\end{aligned}$$

Membership value of Right-Angle triangle, R :

$$\begin{aligned}\mu_R &= 1 - \frac{1}{90^\circ} |X - 90^\circ| \\ &= 1 - \frac{1}{90^\circ} |80^\circ - 90^\circ| \\ &= 1 - \frac{1}{90^\circ} \times 10^\circ = 0.889\end{aligned}$$

Membership value of Equilateral triangle, E :

$$\begin{aligned}\mu_E &= 1 - \frac{1}{180^\circ} (X - Z) \\ &= 1 - \frac{1}{180^\circ} (80^\circ - 45^\circ) \\ &= 1 - \frac{1}{180^\circ} \times 35^\circ = 0.8056\end{aligned}$$

Membership value of Isosceles and Right-Angle Triangle, IR :

$$\begin{aligned}\mu_{IR} &= \min[\mu_I, \mu_R] \\ &= \min[0.833, 0.889] \\ &= 0.833\end{aligned}$$

Membership value of Other Triangles, T :

$$\begin{aligned}\mu_T &= \min[1 - \mu_I, 1 - \mu_E, 1 - \mu_R] \\ &= \min[0.167, 0.1944, 0.111] \\ &= 0.111\end{aligned}$$

$\mu_I = 0.833$
 $\mu_E = 0.8056$
 $\mu_R = 0.889$

Defuzzification

Defuzzification is the process of converting fuzzy output values (which range from 0 to 1) into a single crisp value that can be used in practical applications, such as controlling a device. **Common Defuzzification Methods:**

1. **Centroid Method:** Calculates the centre of gravity of the area under the membership function curve.
2. **Maximum Membership:** Chooses the output value with the highest degree of membership.
3. **Mean of Maxima:** Averages the outputs that have the maximum membership values.

Applications of Fuzzy Sets and Fuzzy Logic

1. **Control Systems:** Used in devices like air conditioners, washing machines, and microwave ovens to manage varying inputs (e.g., temperature, pressure) and adjust output in a human-like manner.
2. **Decision-Making Systems:** Employed in expert systems, decision support systems, and artificial intelligence applications where decisions are based on uncertain or vague information (e.g., medical diagnosis).
3. **Image Processing:** Helps in tasks such as image segmentation, noise reduction, and edge detection, where pixel values may not be well-defined or precise.
4. **Consumer Electronics:** Used in robotics, camera autofocus, and household appliances to interpret vague inputs and provide smoother, more adaptable responses.
5. **Data Classification:** Applied in data mining and machine learning to classify data into categories when exact boundaries are not clear, enhancing performance in systems that need to handle ambiguity.

Advantages of Fuzzy Sets and Fuzzy Logic

1. **Handles Uncertainty and Vagueness:** Fuzzy logic is ideal for dealing with imprecise or incomplete data, representing vague concepts (e.g., "high temperature", "tall person") with degrees of membership.
2. **Human-Like Reasoning:** Mimics human reasoning by dealing with approximate rather than exact reasoning, making systems more intuitive and adaptive.
3. **Flexibility:** Easily integrates into existing systems to improve performance without needing exact mathematical precision, making it highly adaptable for real-world applications.
4. **Simple to Implement:** Fuzzy logic systems are easier to design and implement compared to traditional algorithms for handling uncertainty, especially when working with rule-based systems.
5. **Improved Performance:** Can provide more accurate and efficient decision-making in systems where traditional binary logic would fail, especially in control and automation tasks.

Disadvantages of Fuzzy Sets and Fuzzy Logic

1. **Complexity in Design:** While fuzzy logic is easy to implement in certain applications, designing the rules and membership functions can be complex and time-consuming, especially in large systems.
2. **Lack of Precision:** The results in fuzzy logic are not always precise, as it works with approximate values, which can sometimes lead to less accurate outcomes in some scenarios.
3. **Computational Overhead:** The calculation of fuzzy membership functions and rule evaluation can introduce computational overhead, especially when dealing with large data sets.

4. **Lack of Standardization:** There is no universal framework or standard for defining fuzzy sets and membership functions, leading to inconsistencies across different applications.
5. **Difficulty in Modelling Complex Systems:** For highly intricate or dynamic systems, the design of fuzzy rules may become cumbersome, and it may be challenging to achieve optimal performance.